

Actual-law estimates and two exact singular-tail reductions

HOCF–RIESZ, Round 010

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Scope and provenance. This synthesis records the submitted THM–030 and THM–031 estimates. Their complete proofs are the separately sealed reports from TASK060 and the independent TASK061 falsification route. Fresh review dispositions are recorded separately in the audit and theorem ledgers. Earlier particle, full-inverse and domain statements retain their precise hypotheses. The full expected singular-corrector bracket remains open in this round.

Actual free energy and the two energy floors

Fix $d \geq 3$, $0 < s \leq d - 2$, $T < \infty$, a smooth real terminal test h , and $0 < \nu \leq \nu_*$. Use the unit Haar torus, the frozen coefficient-one Riesz kernel g of mean zero, and $K = -\nabla g$. The actual particles solve

$$dX_i = \frac{1}{N} \sum_{j \neq i} K(X_i - X_j) dt + \sqrt{2\nu} dW_i, \quad N \geq 2,$$

from iid Haar initial positions independent of their Brownian motions. Write $F_N(t)$ for their density and

$$H_N = \frac{1}{N} \sum_{i < j} g(X_i - X_j), \quad \beta = \nu^{-1}, \quad b_N = \min(\beta, 1), \quad \sigma_N^2 = Nb_N.$$

At positive heat cutoff, the genuine smooth gradient dynamics satisfy

$$\frac{d}{dt} [\nu \text{Ent}(F_N^\epsilon) + \mathbb{E}H_N^\epsilon] = - \int F_N^\epsilon |\nabla H_N^\epsilon + \nu \nabla \log F_N^\epsilon|^2 \leq 0.$$

Initial free energy is zero. The fixed- N same-noise path passage, local uniform convergence away from collisions, a common lower energy bound, and weak lower semicontinuity of entropy yield

$$\nu \text{Ent}(F_N(t)) + \mathbb{E}H_N(X_t) \leq 0, \quad \mathbb{E}H_N(X_t) \leq 0.$$

No singular Fisher-information identity is passed. Exchangeability and common translation invariance give a Haar one-body marginal. In particular, with $g_* := \inf g < 0$, the actual pair moment obeys $\mathbb{E}|g(X_1 - X_2)| \leq 2|g_*|$, uniformly in time and parameters.

Set $p = s + 2$ and $a = s/p$. The first construction uses heat convolution and $g_\epsilon \geq g_{\epsilon+r} - \kappa r$ to obtain $H_N^\epsilon \geq -C_0 N^a$. The independent route instead truncates the heat integral. With $\alpha = (d - s)/2$ and $A = 4^{(d-s)/2} \pi^{d/2} / \Gamma(s/2)$, put

$$g^{>r} = A \int_r^\infty t^{\alpha-1} (p_t - 1) dt, \quad a_r(k) = A \int_r^\infty t^{\alpha-1} e^{-4\pi^2 |k|^2 t} dt > 0.$$

Positivity and exact subtraction of the smooth self diagonal give

$$H_N \geq \frac{N}{2} \sum_{k \neq 0} a_r(k) |\hat{\eta}_N(k)|^2 - \frac{1}{2} g^{>r}(0) - \frac{A(N-1)}{2\alpha} r^\alpha, \quad \eta_N = \frac{1}{N} \sum_i \delta_{X_i}.$$

Since $g^{>r}(0) \leq Cr^{-s/2}$, choosing $r = N^{-2/d}$ gives the stronger floor $H_N \geq -CN^{s/d}$. The two powers agree at Coulomb and are compatible throughout $d \geq s + 2$. The cutoff constructions are different objects.

A clipped part of the genuine martingale

Let Φ_t be the genuine symmetric terminal-zero full pair inverse, with both exact compensated responses, and put

$$G_t = \nabla_x \Phi_t, \quad A_t(x) = \int G_t(x, y) dy, \quad H_t = G_t - A_t.$$

Its actual domain theorem supplies the true square-integrable martingale with configuration gradient

$$v_{i,t} = \nabla_i P_N[\Phi_t] = N^{-2} \left[\sum_{j \neq i} H_t(X_i, X_j) - A_t(X_i) \right], \quad Q_N = 2\nu N b_N \mathbb{E} \int_0^T \sum_i |v_{i,t}|^2 dt.$$

Clip G_t radially to length L , obtaining G_t^L , and define its own row mean A_t^L and row-centered field H_t^L . The corresponding v_i^L is bounded and defines an actual martingale. It need not be the gradient of another pair potential: this is an additive decomposition of the genuine martingale integrand.

For $2 \leq k \leq N$, the entropy chain rule, exchangeability and one-body Haar give

$$\text{Ent}(F_N^{(k)}) \leq \frac{k-1}{N-1} \text{Ent}(F_N) \leq \frac{C_0(k-1)N^a}{\nu(N-1)}.$$

The conditional entropy increments increase with the number of conditioning labels; their averages prove the first inequality. Pinsker's inequality in the convention $\|F - 1\|_1 \leq \sqrt{2 \text{Ent}(F)}$ then controls every bounded pair and triple test. Using the exact deleted-label square and the independently supplied Haar energy gives

$$Q_N^L \leq C_E b_N N^{-2/p} + 16T \sqrt{2C_0} b_N \sqrt{\nu} L^2 N^{-1/p}.$$

Here $b_N \sqrt{\nu} = \min(\sqrt{\nu}, \nu^{-1/2}) \leq 1$. Thus for the explicit threshold $L_N = N^{1/(4p)}$,

$$Q_N^{L_N} \leq C_E b_N N^{-2/p} + C_* N^{-1/(2p)} \longrightarrow 0.$$

No evolved iid assumption is used.

Write $G^{>,L} = G - G^L$, with its own row mean and centered field, and let $\mathcal{T}_N(L)$ be the expected scaled bracket of $v - v^L$. The Hilbert norm of a predictable field is the square root of its expected bracket, so

$$\left| \sqrt{Q_N} - \sqrt{\mathcal{T}_N(L)} \right| \leq \sqrt{Q_N^L}.$$

Consequently,

$$Q_N \longrightarrow 0 \iff \mathcal{T}_N(N^{1/(4(s+2))}) \longrightarrow 0. \quad (\text{A})$$

The right-hand assertion is open. For any of these fields, the full quadratic quantity has the exact four-term form

$$\frac{2\nu b_N}{N^2} \int_0^T \left[(N-1) \mathbb{E} |H_{12}|^2 + (N-1)(N-2) \mathbb{E} H_{12} \cdot H_{13} - 2(N-1) \mathbb{E} H_{12} \cdot A_1 + \|A_1\|_2^2 \right] dt.$$

At $N = 2$ the triple object is omitted entirely. The whole expression is nonnegative; separate contractions need not be. No orthogonality of the clipped and tail martingales is asserted.

Actual Fourier control and a smoothing reduction

Put $q = 1 - s/d > 0$. Combining the stronger heat-integral floor with $\mathbb{E}H_N \leq 0$, and using positivity of each Fourier coefficient, yields

$$\mathbb{E}|\hat{\eta}_N(t, k)|^2 \leq \min\{1, CN^{-q}|k|^{d-s}\}, \quad k \neq 0.$$

For $|k| \leq N^{1/d}$, integrate $a_{N^{-2/d}}(k)$ over $[|k|^{-2}, 2|k|^{-2}]$ to obtain a fixed multiple of $|k|^{s-d}$. For higher frequencies use $|\hat{\eta}_N(k)| \leq 1$. The estimate is uniform at deterministic times, including critical sequences; it becomes uninformative at particle-spacing frequencies.

The same floor gives $\text{Ent}(F_N) \leq CN\lambda_N$, where $\lambda_N = \beta_N N^{s/d-1}$. Entropy subadditivity over disjoint fixed-size blocks implies fixed-marginal total-variation convergence when $\lambda_N \rightarrow 0$. At criticality this gives no such convergence. In particular, neither conclusion supplies uniform integrability of singular, N -dependent gradients.

The Borel pair measure inequality and the two finite-measure response bounds give $\sup_{t,N,\nu} \|\Phi_t\|_1 \leq C_h$ by the complete Volterra series. Smooth this already constructed full inverse in both slots:

$$\Psi_{t,\delta} = e^{\delta\Delta_x} e^{\delta\Delta_y} \Phi_t, \quad G_\delta = \nabla_x \Psi_\delta, \quad r_* = (5d - s + 2)/2.$$

Writing m for unit Haar measure, the exact particle derivative is

$$\nabla_i P_N[\Psi_\delta] = \frac{1}{N} \left[\int G_\delta(X_i, y) (\eta_N - m)(dy) - \frac{1}{N} G_\delta(X_i, X_i) \right].$$

Only the smooth diagnostic has this diagonal value. Fourier summation, the actual moment bound, and a supremum in the tagged coordinate give

$$Q_N[\Psi_\delta] \leq C\nu b_N \left[N^{-q}\delta^{-r_*} + N^{-2}\delta^{-(2d+1)} \right] \leq CN^{-q}\delta^{-r_*}.$$

No independence between a tagged particle and the empirical measure is needed. Choose

$$\delta_N = N^{-\gamma}, \quad \gamma = \frac{d-s}{d(5d-s+2)} = \frac{q}{2r_*}.$$

Then $Q_N[\Psi_{\delta_N}] \leq CN^{-q/2} \rightarrow 0$, and the same Hilbert-space triangle inequality proves the second exact equivalence

$$Q_N[\Phi] \rightarrow 0 \iff Q_N[\Phi - \Psi_{\delta_N}] \rightarrow 0. \quad (\text{B})$$

The right side is open. Fixed- N smoothing convergence in Haar H^1 and finite- N density domination do not interchange these limits.

For a real smooth row-centered vector probe H_t , C^1 near time zero, the actual hierarchy also gives the independent sign test, for $N \geq 3$,

$$\left. \frac{d}{dt} \mathbb{E} H_t(X_1, X_2) \cdot H_t(X_1, X_3) \right|_{t=0} = -\frac{2}{N} \int \sum_{k \neq 0} 4\pi^2 c_{d,s} |k|^{s+2-d} |\hat{H}_0(x, k)|^2 dx \leq 0.$$

The two other pair contractions vanish by row centering. Coulomb compensation is retained. This is a fixed-smoothing initial derivative, not a later-time sign or a uniform singular Taylor remainder.

The first open assertion is either equivalent actual-law tail in (A) or (B). The old condition $\beta_N N^{2s/d-1} \rightarrow 0$, full microscopic subcriticality $\lambda_N \rightarrow 0$, and positive finite microscopic criticality remain distinct. No completed fluctuation law or critical hierarchy follows.